

Shardul Dhongade

SF Bay Area, CA

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EDUCATION

Virginia Tech

Masters of Engineering in Computer Science

Blacksburg, VA

August 2024 – May 2025

- Relevant Coursework: Deep Learning, Natural Language Processing, Mobile App Development

Virginia Tech

Bachelors of Science in Computer Science

Blacksburg, VA

August 2021 – May 2024

- President – Association for Computing Machinery (ACM)
- Relevant Coursework: Operating Systems, Data Structures and Algorithms, Machine Learning, Software Design

EXPERIENCE

Associate Software Engineer

Capital One

August 2025 – Present

McLean, VA

- Engineered a React-based troubleshooting banner with dynamic, event-specific hyperlinks to Splunk, AWS Step Functions, and CloudWatch, reducing incident triage search time by 75% (saving up to 1 hour/day)
- Designed and built a high-performance batch retrieval framework from scratch, refactoring a looped API design into a scalable batch architecture
- Built the React/Redux query interface and mapped end-to-end architecture flows, including robust handling for mixed valid/invalid input edge cases
- Wrote extensive Cypress and backend test suites to standardize and launch the endpoint into production, reducing customer query time by 60% and improving API latency by 80%
- Designed a per-intent credential model for a distributed Apache Flink streaming platform, decoupling authentication profiles from enrichment configs via dedicated Auth Profile and Auth Registry Java classes
- Built a type-safe YAML configuration mapping system enabling multiple authentication profiles to be safely parsed into a unified memory map
- Implemented an AWS Secrets Manager-backed credential resolution framework to securely pull, cache, and auto-refresh OAuth2 client credentials, reducing downstream 401 authentication failures

Software Engineer Intern (AI)

Coherent Corp.

June 2025 – August 2025

Santa Clara, CA

- Designed a production-grade, multi-stage agentic AI system using LangChain, OpenAI GPT-4o, FastAPI, and Form.io, enabling natural language-to-structured form conversion for safety incident reporting workflows
- Built a composable LLM agent with function-calling tools and prompt-chained orchestration logic that dynamically adapts based on real-time form state and user input
- Applied a modular Model Context Protocol (MCP) to govern the agent's state transitions, ensuring transparent multi-turn reasoning and error recovery
- Deployed and accelerated Small Language Models (SLMs) on Nvidia Jetson AGX Orin using TensorRT, llama.cpp, and Ollama to enable efficient, low-power LLM inference on edge hardware

Software Engineering Intern

Coherent Corp.

June 2024 – September 2024

Santa Clara, CA

- Designed a Fullstack AI assistant for company ecosystem using OpenAI's Assistant API to provide concurrent visualizations and insights on lab incident data, enhancing decision-making across the company
- Built a responsive, production-grade application using Angular and Bootstrap to expand Health & Safety platform functionalities to over 25 third-party services and contractors

Software Engineering Intern

GEICO

June 2023 – August 2023

Remote

- Created a virtual chatbot for company intranet with Power Virtual Assistant and PowerFx programming, reducing the volume of support tickets by automating responses to common inquiries
- Applied React and Node to set up deployment to SharePoint; configured Azure AD for secure user authentication
- Implemented Agile methodologies with automated workflows, reducing development cycle time by 20%

PROJECTS

Graph Neural Networks for Music Classification | *PyTorch, Jupyter*

- Developed a Graph Neural Network (GNN) using PyTorch and PyTorch Geometric for music genre classification to capture complex relationships between musical features and compare performance to traditional euclidean models
- Implemented node classification through Graph Convolutional Networks (GCNs), leveraging cosine-similarity matrices to construct graph edges and enable transductive learning
- Performed extensive hyperparameter tuning, such as layer depth, neighbor sampling, and regularization methods, optimizing model performance and outperforming some traditional CNN approaches

SkillSprint | *Flask, Python, REST APIs, JavaScript, Next.js, Tailwind*

- Implemented web scraping to extract information from a job posting link, and leveraged OpenAI API and YouTube Data API to categorize job requirements by tech stack and generate tailored study materials
- Built a Flask backend to integrate REST APIs and process data for smooth communication and content delivery
- Designed a responsive Next.js frontend for seamless interaction with Tailwind CSS for an intuitive UI/UX

TECHNICAL SKILLS & CERTIFICATIONS

Programming Languages: Java, Python, JavaScript/TypeScript, C, C++, SQL, R, Kotlin

Frameworks: React.js, Next.js, Angular, Node.js, Flask, FastAPI, MongoDB, Git, Jira

AI & ML: PyTorch, TensorFlow, GCP, Docker, Sklearn, Pandas, NumPy